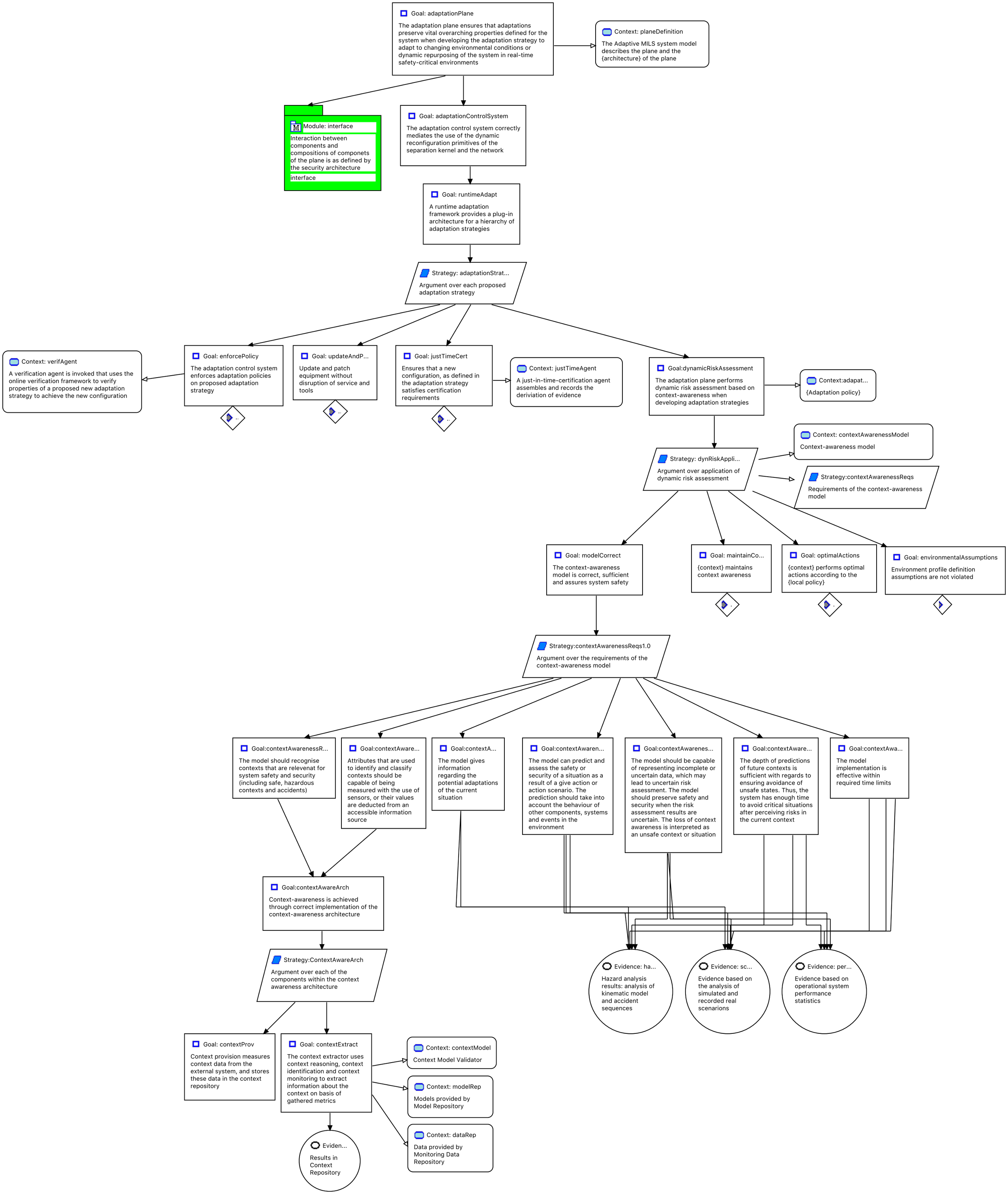
Goal: adaptationPlane
The adaptation plane ensures that adaptations preserve vital overarching properties defined for the system when developing the adaptation strategy to adapt to changing environmental conditions or dynamic repurposing of the system in real-time safety-critical environments
Context: planeDefinition
The Adaptive MILS system model describes the plane and the {architecture} of the plane
Module: interface
Interaction between components and compositions of componets of the plane is as defined by the security architecture
interface
Goal: adaptationControlSystem
The adaptation control system correctly mediates the use of the dynamic reconfiguration primitives of the separation kernel and the network
Goal: runtimeAdapt
A runtime adaptation framework provides a plug-in architecture for a hierarchy of adaptation strategies
Strategy: adaptationStrat...
Argument over each proposed adaptation strategy
Context: verifAgent
A verification agent is invoked that uses the online verification framework to verify properties of a proposed new adaptation strategy to achieve the new configuration
Goal: enforcePolicy
The adaptation control system enforces adaptation policies on proposed adaptation strategy
Goal: updateAndP...
Update and patch equipment without disruption of service and tools
Goal: justTimeCert
Ensures that a new configuration, as defined in the adaptation strategy satisfies certification requirements
Context: justTimeAgent
A just-in-time-certification agent assembles and records the deriviation of evidence
Goal:dynamicRiskAssessment
The adaptation plane performs dynamic risk assessment based on context-awareness when developing adaptation strategies
Context:adapat...
{Adaptation policy}
Context: contextAwarenessModel
Context-awareness model
Strategy: dynRiskAppli...
Argument over application of dynamic risk assessment
Strategy:contextAwarenessReqs
Requirements of the context-awareness model
Goal: modelCorrect
The context-awareness model is correct, sufficient and assures system safety
Goal: maintainCo...
{context} maintains context awareness
Goal: optimalActions
{context} performs optimal actions according to the {local policy}
Goal: environmentalAssumptions
Environment profile definition assumptions are not violated
Strategy:contextAwarenessReqs1.0
Argument over the requirements of the context-awareness model
Goal:contextAwarenessR...
The model should recognise contexts that are relevenat for system safety and security (including safe, hazardous contexts and accidents)
Goal:contextAware...
Attributes that are used to identify and classify contexts should be capable of being measured with the use of sensors, or their values are deducted from an accessible information source
Goal:contextA...
The model gives information regarding the potential adaptations of the current situation
Goal:contextAwaren...
The model can predict and assess the safety or security of a situation as a result of a give action or action scenario. The prediction should take into account the behaviour of other components, systems and events in the environment
Goal:contextAwarenes...
The model should be capable of representing incomplete or uncertain data, which may lead to uncertain risk assessment. The model should preserve safety and security when the risk assessment results are uncertain. The loss of context awareness is interpreted as an unsafe context or situation
Goal:contextAware...
The depth of predictions of future contexts is sufficient with regards to ensuring avoidance of unsafe states. Thus, the system has enough time to avoid critical situations after perceiving risks in the current context
Goal:contextAwa...
The model implementation is effective within required time limits
Goal:contextAwareArch
Context-awareness is achieved through correct implementation of the context-awareness architecture
Strategy:ContextAwareArch
Argument over each of the components within the context awareness architecture
Evidence: ha...
Hazard analysis results: analysis of kinematic model and accident sequences
Evidence: sc...
Evidence based on the analysis of simulated and recorded real scenarions
Evidence: per...
Evidence based on operational system performance statistics
Goal: contextProv
Context provision measures context data from the external system, and stores these data in the context repository
Goal: contextExtract
The context extractor uses context reasoning, context identification and context monitoring to extract information about the context on basis of gathered metrics
Context: contextModel
Context Model Validator
Context: modelRep
Models provided by Model Repository
Context: dataRep
Data provided by Monitoring Data Repository
Eviden...
Results in Context Repository